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REVIEWER SYMBOLS:
NAVY - AS
USAF - 99
ARMY - AV
DLA - IS

USER SYMBOLS:

P.A.
NAVY - AS
Other Coord
USAF - 11
ARMY - AV

PROCUREMENT SPECIFICATION
MIL-B-23964

SUPERSEDES
MS21130(ASC)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES
E 90 KSI Fsu, 63 KSI Ftu, 100° FLUSH HEAD, 450°F 650°F

MILITARY STANDARD
MS21130

SHEET 1 OF 6

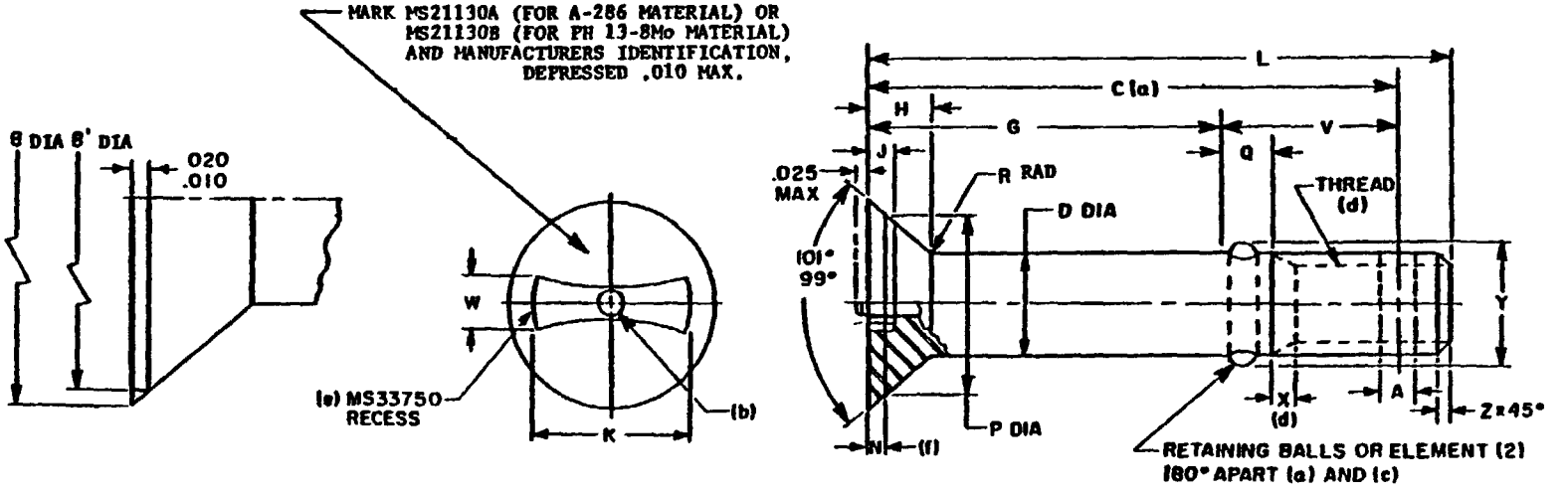
DD FORM 1 MAR 72 672-1 (Coordinated)

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PROJECT NO. 5306-0700 PLATE NO. E

FED. SUP CLASS
5306

(E)



FOR NOTE REQUIREMENTS SEE SHEET 6

(E) - DENOTES CHANGE

REVISED (B) 28 SEP 73 (C) 13 SEP 74 (D) 12 OCT 79 (E) 11 MARCH 1981

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DD FORM 1 MAR 72 672-1 (Coordinated)

PROCUREMENT SPECIFICATION: MIL-B-23964

P.A.: NAVY - AS; Other Cust: USAF - 11, ARMY - AV

SUPERSEDES: MS21130(ASG)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Fcu, 100° FLUSH HEAD, 450°F & 650°F

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

SHEET 2 OF 6

MILITARY STANDARD

MS21130

PLATE NO. 23004

TABLE I

DIAMETER DASH NO.	THREAD DESIGNATION MIL-S-8879	(a) A DIA		B DIA	B' DIA	D DIA		H REF.	J		K	(f) HEAD PROTRUSION N		DIA P		Q (c)		R RAD		V	W	
		MAX.	MIN	MAX	MIN.	MAX.	MIN.		MAX	MIN.	MAX.	MAX.	MIN.	MAX	MIN.	MAX.	MIN.	MAX.	MIN.	BASIC	MAX.	MIN.
3	.1900-32UNJF-3A	.080	.070	.385	.328	.1894	.1889	.080	.057	.054	.270	.0290	.0263	.3147	.3143	.094	.079	.020	.015	293	.098	.094
4	.2500-28UNJF-3A	.086	.076	.507	.449	.2492	.2487	108	.072	.069	.380	.0342	.0316	.4245	.4241	.125	.110			.333	.111	.107
5	.3125-24UNJF-3A			.635	.577	.3117	.3112	.135	.097	.094	.545	.0395	.0370	.5389	.5385	.156	.141	.030	.020	.375	.147	.143
6	.3750-24UNJF-3A			.762	.704	3742	.3737	.162	.111	.107	.630	.0450	.0426	.6532	.6528	.171	.156			.421	.164	.160
7	4375-20UNJF-3A	.116	.106	.890	.832	.4367	.4362	.190	.118	.114	.730	.0503	.0481	.7676	.7672	.202	.187			460	.193	.188
8	.5000-20UNJF-3A			1.017	.959	4991	.4986	.216	.134	.130	.815	.0557	.0537	.8820	.8816	.234	.219			.523	.221	.216
9	.5625-18UNJF-3A	.151	141	1.145	1.087	.5616	.5611	.243	.149	.145	.900	.0611	.0592	.9964	.9960	.265	.250			.628	.262	.257

(a) GENERAL DIMENSIONAL AND MECHANICAL REQUIREMENTS:
(E) COTTER PIN HOLE SHALL BE WITHIN .010 INCH OF BOLT CENTERLINE FOR BOLTS .3125 DIA. AND SMALLER AND WITHIN .015 INCH FOR BOLTS .3750 DIA AND LARGER.
(E) THE PIN HOLE SHALL BE PERPENDICULAR TO BOLT CENTERLINE. LOCATION OF COTTER PIN HOLE FROM TOP OF HEAD IS GIVEN BY C=(G MIN.+ V) ±.010.
THE COTTER PIN HOLE SHALL BE PERPENDICULAR TO THE SLOT IN HEAD WITHIN 5°. ORIENTATION OF BALLS OR ELEMENT TO COTTER PIN HOLE SHALL BE IN LINE WITHIN 3°. BALL OR ELEMENT PROTRUSION FROM THE BOLT CENTERLINE SHALL BE EQUAL OR LESS THAN 1/2 Y MAX.

(b) (E) THE RETAINING DEVICE SHALL BE RELEASED ONLY BY DEPRESSING THE PLUNGER LOCATED IN THE HEAD OF THE BOLT. MAX DIAMETER OF PLUNGER RECESS HEAD SHALL BE THE MINIMUM VALUE FOR THE "M" DIMENSIONS ON MS33750.

DIAMETER DASH NO.	Y		Z		(e) RECESS NO.
	MAX.	MIN	MAX.	MIN.	
3	.225	.215	.041	.021	2
4	.294	.284			3
5	.380	.370	.057	.037	4
6	.445	.435			5
7	.514	.504			6
8	.599	.589			7
9	.671	.661	072	.052	8

(c) BALLS OR RETAINING ELEMENTS SHALL BE LOCATED WITHIN THE LIMITS OF Q DIMENSION AND AT MAXIMUM PROTRUSION SHALL FIT WITHIN THE GROOVES OF THE APPROPRIATE SPACER OF MS21126 AND MS21128
E WITHOUT INTERFERENCE OR BINDING.

(d) THREAD: IN ACCORDANCE WITH MIL-S-8879 EXCEPT THAT INCOMPLETE THREAD LENGTH (X) SHALL BE A MAXIMUM OF 1-1/2 AND A MINIMUM OF 1/2 THREAD PITCHES IN LENGTH AND MAJOR DIA. TO BE A MIN. OF .001 LESS THAN MEASURED SHANK DIA.

(e) RECESS SHALL BE LOCATED AT TRUE POSITION TO "B" WITHIN .010 AND SHALL CONFORM TO MS33750 EXCEPT THAT RECESS DIMPLE IS OPTIONAL.

(f) REFERENCE NAS518 AND NAS519 FOR FLUSH GAGING DETAILS.

(g) CONCENTRICITY: D AND B TO BE CONCENTRIC TO P.D. OF THREAD WITHIN .002 TIR AT MAXIMUM MATERIAL CONDITION.

FED. SUP CLASS 5306

APPROVED 15 DEC 67 REVISED (E) FOR CHANGES SEE SHEETS 1, 2 and 6

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PROCUREMENT SPECIFICATION
MIL-B-23964

SUPERSEDES
MS 21130 (ASG)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Ftu, 100° FLUSH HEAD, 450°F & 650°F

MILITARY STANDARD
MS21130

SHEET 3 OF 6

DD FORM 1426 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 1532

FED. SUP CLASS 5306

TABLE II

GRIP DASH NO:	NOMINAL SIZE / THREAD DESIGNATION / GRIP LENGTH		.1900 (10) 32 UNJF-3A L-LENGTH		.2500 (1/4) 28 UNJF-3A L-LENGTH		.3125 (5/16) 24 UNJF-3A L-LENGTH		.3750 (3/8) 24 UNJF-3A L-LENGTH		.4375 (7/16) 20 UNJF-3A L-LENGTH		.5000 (1/2) 20 UNJF-3A L-LENGTH		.5625 (9/16) 18 UNJF-3A L-LENGTH		GRIP DASH NO.
	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
03	.204	.188	580	.560	.626	.606											03
04	.266	.250	.642	.622	.688	.668	.748	.728	.809	.789							04
05	.328	.312	.705	.685	.751	.731	.810	.790	.871	.851							05
06	.390	.375	.767	.747	.813	.793	.873	.853	.934	.914	.978	.958	1.041	1.021			06
07	.454	.438	.830	.810	.876	.856	.935	.915	.996	.976	1.040	1.020	1.103	1.083	1.244	1.224	07
08	.516	.500	.892	.872	.938	.918	.998	.978	1.059	1.039	1.103	1.083	1.166	1.146	1.306	1.286	08
09	.578	.562	955	.935	1.001	.981	1.060	1.040	1.121	1.101	1.165	1.145	1.228	1.208	1.369	1.349	09
10	.641	.625	1.017	.997	1.063	1.043	1.123	1.103	1.184	1.164	1.228	1.208	1.291	1.271	1.431	1.411	10
11	.704	.688	1 080	1.060	1.126	1.106	1.185	1.165	1.246	1.226	1.290	1.270	1.353	1.333	1.494	1.474	11
12	.766	.750	1.142	1.122	1.188	1.168	1.248	1.228	1.309	1.289	1.353	1.333	1.416	1.396	1.556	1.536	12
13	.828	.812	1.205	1 185	1.251	1.231	1.310	1.290	1.371	1.351	1.415	1.395	1.478	1.459	1.619	1 599	13
14	.891	.875	1.267	1.247	1.313	1.293	1.373	1.353	1.434	1.414	1.478	1.458	1.541	1.521	1.681	1.661	14
15	.954	.938	1.330	1.310	1.376	1.356	1.435	1.415	1.496	1.476	1.540	1.520	1.603	1.583	1.744	1.724	15
16	1.016	1.000	1.392	1.372	1.438	1.418	1.498	1.478	1.559	1.539	1.603	1.583	1.666	1.646	1.806	1.786	16
17	1.078	1.062	1.455	1.435	1.501	1.481	1.560	1.540	1.621	1.601	1.665	1.645	1.728	1.708	1.869	1.849	17
18	1.141	1.125	1.517	1.497	1.563	1.543	1.623	1.603	1.684	1.664	1.728	1.708	1.791	1.771	1.931	1.911	18
19	1.203	1.188	1.580	1.560	1.626	1.606	1.685	1.665	1.746	1.726	1.790	1.770	1.853	1.833	1.994	1.974	19
20	1.266	1.250	1 642	1.622	1.688	1.668	1.748	1.728	1.809	1.789	1.853	1.833	1.916	1.896	2.056	2.036	20
21	1.328	1.312	1.705	1.685	1.751	1.731	1.810	1.790	1.871	1.851	1.915	1.895	1.978	1 958	2.119	2 099	21
22	1.391	1.375	1.767	1.747	1.813	1.793	1.873	1.853	1.934	1.914	1.978	1.958	2.041	2.021	2.181	2.161	22
23	1.454	1.438	1.830	1.810	1.876	1.856	1.935	1.915	1.998	1.976	2.040	2.020	2.103	2.083	2.244	2.224	23
24	1.516	1.500	1.892	1.872	1.938	1.918	1.998	1.978	2.059	2.039	2.103	2 083	2.166	2.146	2.306	2.286	24
25	1.578	1.562	1.954	1.934	2.001	1.981	2.060	2.040	2.121	2.101	2.165	2.145	2.228	2.208	2.369	2.349	25
26	1.641	1.625	2.017	1.997	2.063	2.043	2.123	2 103	2.184	2.164	2.228	2.208	2.291	2.271	2.431	2.411	26
27	1.704	1.688	2.080	2.060	2.126	2.106	2.185	2.165	2.246	2.226	2.290	2.270	2.353	2.333	2.494	2.474	27
28	1.766	1.750	2.142	2.122	2.188	2.168	2.248	2.228	2.309	2.289	2.353	2.333	2.416	2.396	2.556	2.536	28
29	1.828	1.812	2.205	2.185	2.251	2.231	2.310	2.290	2.371	2.351	2.415	2.395	2.478	2.458	2.619	2.599	29
30	1.891	1.875	2.267	2.247	2.313	2.293	2.373	2.353	2.434	2.414	2.478	2.458	2.541	2.521	2.681	2.661	30
31	1.953	1.937	2.330	2.310	2.376	2.356	2.435	2.415	2.496	2.476	2.540	2.520	2.603	2.583	2.744	2.724	31
32	2.016	2.000	2.392	2.372	2.438	2.418	2.498	2.478	2.559	2.539	2.603	2.583	2.666	2.646	2.806	2.786	32
33	2.078	2.062	2.455	2.435	2.501	2.481	2.560	2.540	2.621	2.601	2.665	2.645	2.728	2.708	2.869	2.849	33
34	2.141	2.125	2.517	2.497	2.563	2.543	2.623	2.603	2.684	2.664	2.728	2.708	2.791	2.771	2.931	2.911	34

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DD FORM 1 MAR 72 672-1 (Coordinated)

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Other Cust
USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-23864

SUPERSEDES
MS21130 (ASG)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES
90 KSI Fsu, 63 KSI Ftu, 100° FLUSH HEAD, 450°F & 650°F

MILITARY STANDARD
MS21130

SHEET 4 OF 6

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PLATE NO 17709

FED. SUP CLASS
5306

TABLE III

GRIP DASH NO.	NOMINAL SIZE		.1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
35	2.204	2.188	2.580	2.560	2.626	2.606	2.685	2.665	2.746	2.726	2.790	2.770	2.853	2.833	2.994	2.974	35
36	2 266	2.250	2.642	2.622	2.688	2.668	2.748	2.728	2.809	2.789	2 853	2.833	2.916	2.896	3 056	3.036	36
37	2.328	2.312	2.705	2.685	2.751	2.731	2.810	2.790	2.871	2.851	2.915	2.895	2.978	2.958	3.119	3.099	37
38	2.390	2.375	2.767	2.747	2.813	2.793	2.873	2.853	2 934	2.914	2.978	2.958	3.041	3.021	3.181	3.161	38
39	2.454	2.438	2.830	2.810	2.876	2.856	2.935	2.915	2.996	2.976	3 040	3.020	3.103	3.083	3.244	3.224	39
40	2.516	2.500	2.892	2.872	2.938	2.918	2.998	2.978	3.059	3.039	3.103	3.083	3.166	3.146	3.306	3.286	40
41	2.578	2.562	2.954	2.934	3.001	2.981	3.060	3 040	3.121	3.101	3.165	3.145	3.228	3.208	3.369	3.349	41
42	2.641	2.625	3.017	2.997	3.063	3.043	3.123	3 103	3.184	3.164	3.228	3.208	3.291	3.271	3.431	3.411	42
43	2.704	2 688	3.080	3.060	3.126	3 106	3.185	3.165	3.246	3.226	3.290	3.270	3.353	3.333	3.494	3.474	43
44	2.766	2 750	3.142	3.122	3.188	3.168	3.248	3.228	3.309	3.289	3 353	3.333	3.416	3.396	3.556	3.536	44
45	2.828	2.812	3.205	3.185	3.251	3.231	3.310	3.290	3.371	3.351	3.415	3.395	3.478	3 458	3 619	3.599	45
46	2.891	2.875	3.267	3.247	3.313	3.293	3.373	3.353	3.434	3.414	3.478	3 458	3.541	3.521	3.681	3.661	46
47	2.954	2.938	3.330	3.310	3.376	3.356	3.435	3.415	3.496	3.476	3.540	3.520	3.603	3.583	3 744	3 724	47
48	3.016	3 000	3.392	3.372	3.438	3.418	3.498	3.478	3.559	3.539	3.603	3.583	3.666	3.646	3.806	3.786	48
49	3.078	3.062	3.455	3.435	3.501	3.481	3.560	3.540	3.621	3.601	3.665	3.645	3.728	3.708	3.869	3.849	49
50	3.141	3.125	3.517	3.497	3 563	3.543	3.623	3.603	3.684	3.664	3.728	3.708	3.791	3.771	3.931	3.911	50
51	3.203	3.188	3.580	3.560	3.626	3.606	3.685	3.665	3.746	3.726	3.790	3.770	3.853	3.833	3.994	3.974	51
52	3.266	3.250	3 642	3.622	3.688	3.668	3.748	3 728	3.809	3.789	3.853	3.833	3.916	3.896	4 056	4.036	52
53	3.328	3.312	3.705	3.685	3.751	3.731	3.810	3.790	3.871	3.851	3.915	3.895	3.978	3.958	4.119	4.099	53
54	3.391	3.375	3.767	3.747	3.813	3.793	3.873	3.853	3.934	3.914	3.978	3.958	4.041	4.021	4.181	4 161	54
55	3 454	3.438	3.830	3.810	3.876	3,856	3.935	3.915	3.996	3.976	4.040	4.020	4.103	4.083	4.244	4.224	55
56	3.516	3.500	3.892	3.872	3.938	3.918	3.998	3.978	4.059	4.039	4.103	4.083	4.166	4.146	4.306	4.286	56
57	3.578	3.562	3 954	3.934	4.001	3.981	4.060	4.040	4.121	4.101	4 165	4.145	4.228	4.208	4.369	4.349	57
58	3 641	3 625	4 017	3.997	4.063	4.043	4.123	4.103	4.184	4.164	4.228	4.208	4.291	4 271	4.431	4 411	58
59	3 704	3.688	4.080	4.060	4.126	4.106	4.185	4.165	4.246	4.226	4.290	4.270	4.353	4.333	4.494	4 474	59
60	3.766	3 750	4.142	4.122	4.188	4.168	4.248	4.228	4.309	4.289	4.353	4.333	4.416	4.396	4.556	4.536	60
61	3.828	3.812	4.205	4.185	4.251	4.231	4.310	4.290	4.371	4.351	4 415	4.395	4.478	4.458	4.619	4 599	61

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Other Cust
USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-23904

SUPERSEDES
MS21130 (ASG)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES
90 KSI Fsu, 63 KSI Fsu, 100° FLUSH HEAD, 450°F & 650°F

MILITARY STANDARD
MS21130

SHEET 5 OF 6

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 17769

FED. SUP CLASS
5306

TABLE IX

GRIP DASH NO.	NOMINAL SIZE		.1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
62	3.891	3.875	4.267	4.247	4.313	4.293	4.373	4.353	4.434	4.414	4.478	4.458	4.541	4.521	4.681	4.661	62
63	3.953	3.937	4.330	4.310	4.376	4.356	4.435	4.415	4.496	4.476	4.540	4.520	4.603	4.583	4.744	4.724	63
64	4.016	4.000	4.392	4.372	4.438	4.418	4.498	4.478	4.559	4.539	4.603	4.583	4.666	4.646	4.806	4.786	64
65	4.078	4.062	4.455	4.435	4.501	4.481	4.560	4.540	4.621	4.601	4.665	4.645	4.728	4.708	4.869	4.849	65
66	4.141	4.125	4.517	4.497	4.563	4.543	4.623	4.603	4.684	4.664	4.728	4.708	4.791	4.771	4.931	4.911	66
67	4.203	4.188	4.580	4.560	4.626	4.606	4.685	4.665	4.746	4.726	4.790	4.770	4.853	4.833	4.994	4.974	67
68	4.266	4.250	4.642	4.622	4.688	4.668	4.748	4.728	4.809	4.789	4.853	4.833	4.916	4.896	5.056	5.036	68
69	4.328	4.312	4.705	4.685	4.751	4.731	4.810	4.790	4.871	4.851	4.915	4.895	4.978	4.958	5.119	5.099	69
70	4.391	4.375	4.767	4.747	4.813	4.793	4.873	4.853	4.934	4.914	4.978	4.958	5.041	5.021	5.181	5.161	70
71	4.454	4.438	4.830	4.810	4.876	4.856	4.935	4.915	4.996	4.976	5.040	5.020	5.103	5.083	5.244	5.224	71
72	4.516	4.500	4.892	4.872	4.938	4.918	4.998	4.978	5.059	5.039	5.103	5.083	5.166	5.146	5.306	5.286	72
73	4.578	4.562	4.954	4.934	5.001	4.981	5.060	5.040	5.121	5.101	5.165	5.145	5.228	5.208	5.369	5.349	73
74	4.641	4.625	5.017	4.997	5.063	5.043	5.123	5.103	5.184	5.164	5.228	5.208	5.291	5.271	5.431	5.411	74
75	4.704	4.688	5.080	5.060	5.126	5.106	5.185	5.165	5.246	5.226	5.290	5.270	5.353	5.333	5.494	5.474	75
76	4.766	4.750	5.142	5.122	5.188	5.168	5.248	5.228	5.309	5.289	5.353	5.333	5.416	5.396	5.556	5.536	76
77	4.828	4.812	5.205	5.185	5.251	5.231	5.310	5.290	5.371	5.351	5.415	5.395	5.478	5.458	5.619	5.599	77
78	4.891	4.875	5.267	5.247	5.313	5.293	5.373	5.353	5.434	5.414	5.478	5.458	5.541	5.521	5.681	5.661	78
79	4.953	4.937	5.330	5.310	5.376	5.356	5.435	5.415	5.496	5.476	5.540	5.520	5.603	5.583	5.744	5.724	79
80	5.016	5.000	5.392	5.372	5.438	5.418	5.498	5.478	5.559	5.539	5.603	5.583	5.666	5.646	5.806	5.786	80
81	5.078	5.062	5.455	5.435	5.501	5.481	5.560	5.540	5.621	5.601	5.665	5.645	5.728	5.708	5.869	5.849	81
82	5.141	5.125	5.517	5.497	5.563	5.543	5.623	5.603	5.684	5.664	5.728	5.708	5.791	5.771	5.931	5.911	82
83	5.203	5.188	5.580	5.560	5.626	5.606	5.685	5.665	5.746	5.726	5.790	5.770	5.853	5.833	5.994	5.974	83
84	5.266	5.250	5.642	5.622	5.688	5.668	5.748	5.728	5.809	5.789	5.853	5.833	5.916	5.896	6.056	6.036	84
85	5.328	5.312	5.705	5.685	5.751	5.731	5.810	5.790	5.871	5.851	5.915	5.895	5.978	5.958	6.119	6.099	85
86	5.390	5.375	5.767	5.747	5.813	5.793	5.873	5.853	5.934	5.914	5.978	5.958	6.041	6.021	6.181	6.161	86

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5306

TABLE V

SIZE	SHANK DOUBLE SHEAR ULTIMATE STRENGTH LBS MIN	BOLT AND NUT ASSEMBLY ULTIMATE TENSILE STRENGTH LBS MIN SEE NOTE 13	RETAINING DEVICE ULTIMATE TENSILE LOAD LBS MIN
.1900	5,150	1,110	110
.2500	9,200	2,040	230
.3125	14,400	3,250	510
.3750	20,600	5,050	575
.4375	28,000	6,800	710
.5000	36,800	9,250	1,160
.5625	46,000	11,800	1,420

NOTES:

1. MATERIALS, MAGNETIC: BOLT AND PLUNGER: CORROSION RESISTING STEEL PH 13-8Mo CHEMISTRY OF AMS5629 (E)
   BALLS OR ELEMENT: CORROSION RESISTING STEEL PER QQ-S-763 CLASS FSC 4400
   SPRING: CORROSION RESISTING STEEL WIRE, SPRING TEMPER PER AMS 5673, 17-7 PH OR QQ-W-423, CLASS FS302, CONDITION B.
2. HEAT TREATMENT: BOLT AND PLUNGER: PER MIL-H-6875 TO CONDITION H1000 RC 43 TO 47 (E)
   BALLS: PER MIL-H-6875 TO ROCKWELL C50 TO C62, 255 KSI (E) (E)
3. MATERIALS, NONMAGNETIC: BOLT AND PLUNGER: CORROSION RESISTING STEEL A-286 CHEMISTRY OF AMS 5737
   (E) BALLS OR ELEMENT: CORROSION RESISTING STEEL PER MIL-S-7720 18-^8 MARGINAL FOR MAG. PERM.
   SPRING: NICKEL-CHROMIUM-IRON ALLOY PER QQ-W-390 CONDITION C OR NICKEL-ALLOY SPRING PER AMS 5699, SPRING TEMPER
4. HEAT TREATMENT: BALLS: PER MIL-H-6875 TO ROCKWELL C25 MINIMUM 120 KSI (E)
5. PROTECTIVE TREATMENT: CRES MATERIAL PASSIVATED PER QQ-P-35
6. SURFACE TEXTURE: UNTHREADED SHANK AND BEARING SURFACE UNDER HEAD SHALL BE 32 MICROINCHES, ALL OTHER SURFACES SHALL BE 125 MICROINCHES IN ACCORDANCE WITH ANSI B46.1.
7. BOLTS SHALL BE FREE FROM BURRS AND SLIVERS.
8. EXAMPLE OF PART NUMBER:

   MS21130A 4 16 = .2500 UNJF-3A BOLT, 1 INCH NOMINAL GRIP, SLOTTED HEAD A-286 MATERIAL (E)

   - 4 — SEE TABLES I, II, III AND IV FOR DASH NUMBERS
   - 4 — GRIP DASH NUMBER
   - 16 — DIAMETER DASH NUMBER
   - A — MATERIAL - "A" FOR A-286 OR "B" FOR PH 13-8Mo (E)
   - MS21130 — MS NUMBER
9. DESIGN USAGE: FOR DESIGN AND USAGE LIMITATIONS, SEE MS33602.
10. (E) MAGNETIC PERMEABILITY: FOR BOLT ASSEMBLY WITH BOLT AND PLUNGER MADE FROM A-286 MATERIAL. MAGNETIC PERMEABILITY SHALL BE LESS THAN 2.0 (AIR-1.0) FOR A FIELD STRENGTH =200 ORSTEADS USING MAGNETIC PERMEABILITY INDICATOR PER MIL-I-17214 OR EQUIVALENT.
11. TEMPERATURE LIMITATION: 450°F MAX. FOR PH 13-8Mo AND 650°F MAX FOR A-286 MATERIAL.
12. (E) INTERCHANGEABILITY RELATIONSHIP: BOLTS MADE FROM PH17-4 (CODE "-") ARE INACTIVE FOR NEW DESIGN AND AND CANCELLED AFTER 15 MAY 1981.
    BOLTS MADE FROM CRES PH13-8Mo (CODE "B") MAY BE SUBSTITUTED FOR BOLTS MADE FROM PH17-4 IN ALL APPLICATIONS.
    BOLTS MADE FROM PH17-4 (CODE "-") SHALL NOT BE USED IN PLACE OF BOLTS MADE FROM CRES PH13-8Mo.
    BOLTS MADE FROM CRES PH 17-4 SHALL NOT BE USED AFTER 1 JAN 1983.
13. TENSILE LOAD IS BASED ON THE AREA AT MINOR DIAMETER IN FED-STD-H28/2 AND 62.5 KSI Ftu.
14. BOLT PLUNGER MANUFACTURED FROM PH 13-8Mo AND A-286 MATERIAL SHALL BE COATED WITH DRY FILM LUBRICANT PER MIL-L-8937 AND MIL-L-46010 RESPECTIVELY.
15. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
16. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 15 DEC 67 REVISED (E) FOR CHANGES SEE SHEETS 1, 2 & 6

P.A. NAVY - AS Other Cust: USAF -11 ARMY -AV	TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Ftu, 100° FLUSH HEAD, 450°F & 650°F	MILITARY STANDARD MS21130
PROCUREMENT SPECIFICATION MIL-B-23964	SUPERSEDES MS21130(ASG)	SHEET 6 OF 6

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